

code, with examples of function overloading.

```
int sum(int x, int y);
float sum(float x, float y);
int sum(int x, int y, int z);
```

Write two simple C++ programs with definitions of the above functions in *fun.cpp*, the main function with a few calls in *test.cpp*, and then generate object files.

```
g++ -c fun.cpp
g++ -c test.cpp
```

Now check the symbols generated for function definitions and calls for the sum and swap functions using *nm*.

```
nm test.o fun.o
```

Here, you can see overloaded function calls. Definitions are decorated according to the number and type of parameters, and the enclosing scope like global, class, namespace, usage of templates, etc. But names are mangled as per internal conventions. To demangle the symbol names in readable format, you can use the option *-C* or *--demangle* with *nm* or *objdump*.

```
nm --demangle test.o fun.o
objdump -t -C test.o fun.o
```

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Some advanced examples

Using *moment.js* inside Nashorn: *moment.js* is a pretty lightweight and wonderful isomorphic JavaScript library that makes things like *DateTime* formatting, date localisation and *DateTime* arithmetic work like a breeze. Any Java developer would agree that formatting dates in Java with multiple try-catch blocks is pretty messy, and would appreciate the ease *moment.js* brings to the task.

To get started, download *moment.js* from <http://momentjs.com/downloads/moment-with-locales.js> and dump it into your *Resources* folder. The code *eval1* for evaluating in Java Runtime can be downloaded from http://opensourceforu.com/article_source_code/nov16/nashorn.zip.

Writing *react.js/es6* syntax and transpiling it with Babel in Nashorn: As mentioned earlier, Nashorn currently supports only ES 5.1 syntax, and you need to run it through a transpiler like Babel before executing it on the Nashorn engine.

The good thing about Babel is that it comes in a JavaScript standalone version (which can be downloaded from <https://cdnjs.cloudflare.com/ajax/libs/babel-core/6.1.7/browser.min.js>). It's the complete Babel library

A note about *elfutils* and cross toolchains

GNU tools target a wide variety of platforms. Ulrich Drepper wrote *elfutils* purely for Linux and the ELF format. You can download these from the package manager or build them from source available at *fedorahosted.org/releases/e/elfutils*. Similar tools are also available for various target platforms with suitable prefixes, e.g., the Linaro family of toolchains for building Linux components, and bare metal toolchains from *launchpad.net/gcc-arm-embedded* for ARM targets provide similar tools with their own prefixes. The options and usage of these tools will be similar to the steps mentioned here.

In this article, only minimal hints on tools have been provided. Please refer to the man pages of each command and other resources for additional inputs. 

References

- [1] gcc.gnu.org, gnu.org/s/binutils
- [2] wikipedia.org/wiki/GNU_Binutils
- [3] mingw.org, mingw-w64.org
- [4] *Man pages*

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in a single JS file, which can be evaluated on Nashorn and then be re-used to transpile ES6 JS into ES 5.1 JavaScript in the code.

Download the library, dump it into the *Resources* folder of the project, and use the *trans1* from the *Nashorn.zip* downloaded from the above link to see the translation at runtime.

This article just covers the tip of the iceberg as you can do a lot more with Nashorn like creating UIs with *JavaFX*, rendering HTML mark-up on the server side, etc. You can find examples for more scenarios (along with the above mentioned examples) at <https://github.com/shivasaxena/java-nashorn-examples>. 

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